Tutorial 05

Expected Value, (Co)Variance, Independence and Uncorrelatedness, Joint and Marginal pdf

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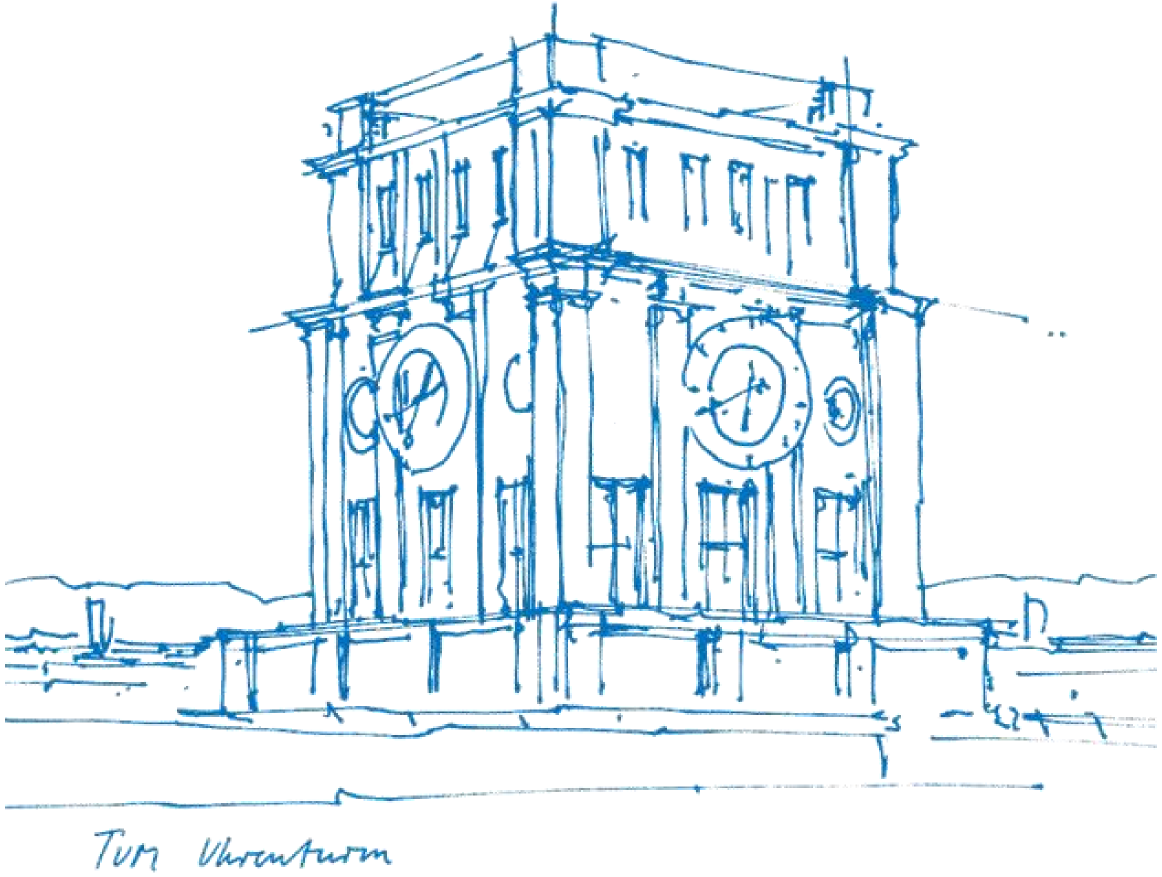

About Tutorial

- **Tutorial:** In-Person, 2h by two time periods per week - 12:00-14:00 Tue, 03.11.018 (Group B33)
  - 10:00-12:00 Fri, 01.06.011 (Group E24)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code ☞

https://drive.google.com/drive/folders1rBeL0NKHuYzrULK3Fr4CGr8svRhyTNOO?usp=sharing

➢ PART I. Recap of the Lecture

➢ PART II. Exercises from the Exercise Sheet

➢ PART III. Additional Exercises (with solution)

I. Recap

- **Expected Value, Variance and Covariance**
  - ① - Definitions
  - ② - Properties
- **Joint and Marginal pdf**
  - ⑤ - Definitions
- **Independece and Uncorrelatedness**
  - ③ - Independece
    - Definition
    - Implication
  - ④ - Uncorrelatedness
    - Definition

Distribution	Definition	Expectation	Variance
Bernoulli	$f_X(x) = \begin{cases} p & x = 1 \\ 1-p & x = 0 \end{cases}$	$\mathbb{E}[X] = p$	$\mathbb{Var}[X] = p(1-p)$
Binomial	$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$	$\mathbb{E}[X] = np$	$\mathbb{Var}[X] = np(1-p)$
Geometric	$f_X(i) = p(1-p)^{i-1}$	$\mathbb{E}[X] = \frac{1}{p}$	$\mathbb{Var}[X] = \frac{1-p}{p^2}$
Poisson	$f_X(i) = \frac{e^{-\lambda}\lambda^i}{i!}$	$\mathbb{E}[X] = \lambda$	$\mathbb{Var}[X] = \lambda$

II. Exercise Sheet

Exercise 1. Let $(\Omega, \mathcal{A}, P)$ be a probability space and $X, Y : \Omega \to \mathbb{R}$, $Z_1, Z_2, Z_3 : \Omega \to \mathbb{R}^n$ random variables/vectors and $a, b \in \mathbb{R}$. Verify the following computation rules for variance and covariance by using their definitions and the properties of the expectation we have mentioned in the lecture:

$$\operatorname{var}[aX + bY] = a^2 \operatorname{var}[X] + b^2 \operatorname{var}[Y] + 2ab \operatorname{cov}[X, Y]$$

and

$$\operatorname{cov}[aZ_1 + bZ_2, Z_3] = a \operatorname{cov}[Z_1, Z_3] + b \operatorname{cov}[Z_2, Z_3] .$$

[Hint: You can use that for random vectors Z_1, Z_2 we have $\operatorname{cov}[Z_1, Z_2] = \mathbb{E}\left[(Z_1 - \mathbb{E}[Z_1])^\top (Z_2 - \mathbb{E}[Z_2])\right]$, where $Z_1^\top$ is the transpose of Z_1.

II. Exercise Sheet

Exercise 2. Calculate the mean and variance for

(i) the discrete RV $X : \Omega \to \mathbb{N}$ with pmf $\rho(n) = \frac{\lambda^n}{n!} e^{-\lambda}$ for a fixed $\lambda > 0$.

(ii) the binomially distributed RV $X : \Omega \to \{0, \ldots, n\}$, meaning X models the number of successes observed in $n \in \mathbb{N}$ i.i.d trials where each trial is Bernoulli distributed with probability of success $p \in (0, 1)$.

[Hint: Consider writing $X = \sum_{i=1}^{n} Y_i$ for $Y_i \overset{\text{iid}}{\sim} \text{Ber}(n, p)$ and use Exercise 1.]

Exercise 3. Let X and Y be two jointly continuous random variables with joint pdf

$$p_{X,Y}(x,y) = \begin{cases} \frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6} & \text{if } x \in [0,1],\ y \in [0,2]\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

For $y \in [0,2]$, find

(i) the conditional pdf of X given $Y = y$, i.e., the function $p_{X|Y}(x \mid y)$,

(ii) the probability $P(X < \frac{1}{2} \mid Y = y)$.

II. Exercise Sheet

Exercise 4. Let $(\Omega, \mathcal{A}, P)$ be a probability space and $X, Y : \Omega \to \mathbb{R}$ be random variables. Then show the following.

(i) If X and Y are independent, they are uncorrelated.

(ii) Give an example for X and Y being uncorrelated but not independent.
[Hint: Choose $Y = X^2$ for an appropriate choice of RV X. To show the non-independence, work with the cdfs.]

III. Additional Exersices

Exercise 5. In a test, the DPT tutor Angelika gives her students three tasks. For each correctly solved task, Angelika awards one point. If a task is solved incorrectly, no point is given for that task. Assume that the students earn points randomly in the test. Let X_i be the score from the i-th task, and let

$$f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \begin{cases} (x_1+x_2+x_3-1)^2/8 & \text{for } x_1,x_2,x_3 \in \{0,1\} \\ 0 & \text{otherwise} \end{cases}$$

be the joint density.

(a) Determine the marginal densities f_{X_1}, f_{X_2}, and f_{X_3}.

(b) Are the random variables X_i independent?

(c) Calculate the expected value and the variance of the total score.

(a) $$f_{X_1} = \sum_{x_2 \in W_{X_2}} \sum_{x_3 \in W_{X_3}} f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \begin{cases} \frac{1}{4} & x_1=0 \\ \frac{3}{4} & x_1=1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_{X_2} = \begin{cases} \frac{1}{4} & x_2=0 \\ \frac{3}{4} & x_2=1 \\ 0 & \text{otherwise} \end{cases} \qquad f_{X_3} = \begin{cases} \frac{1}{4} & x_3=0 \\ \frac{3}{4} & x_3=1 \\ 0 & \text{otherwise} \end{cases}$$

(b) To proof: $f_{X_1,X_2 \cdots X_n}(x_1, \cdots x_n) = f_{X_1}(x_1) \cdot f_{X_2}(x_2) \cdot \cdots \cdot f_{X_n}(x_n)$

$$f_{X_1}(x_1) \cdot f_{X_2}(x_2) \cdot f_{X_3}(x_3) = \frac{2x_1^2+1}{4} \cdot \frac{2x_2^2+1}{4} \cdot \frac{2x_3^2+1}{4}.$$

(All the results plug in)

$$f_{X_1,X_2,X_3}(0,0,0) = \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64} \neq \frac{1}{8}$$

(c) $Y = X_1 + X_2 + X_3.$ $\quad (Y^2 = (X_1+X_2+X_3)^2)$

$$E[Y] = E[X_1] + E[X_2] + E[X_3] = \left(0 \cdot \frac{1}{4} + 1 \cdot \frac{3}{4}\right) \cdot 3 = \frac{9}{4}.$$

$$\mathrm{Var}[Y] = E[Y^2] - E[Y]^2 = 0 \times \frac{1}{8} \times 1 + 1 \times \frac{0}{8} \times 3 + 4 \times \frac{1}{8} \times 3 + 9 \times \frac{4}{8} \times 1 - \left(\frac{9}{4}\right)^2$$

$$= \frac{3}{2} + \frac{9}{2} - \frac{81}{16} = 6 - \frac{81}{16} = \frac{15}{16}.$$